trillion with ends was far too few to be discerned in any nineteenth-century physics experiment, the fact that Maxwell's laws predicted such a thing seemed rather complicated and ugly to Lorentz, Poincaré, and Larmor.

Now, the Newtonian laws of physics were unequivocal: Time is *absolute.* It flows uniformly and inexorably at the same universal rate, independently of how one moves. If the Newtonian laws were correct, then motion cannot cause time to dilate any more than it can cause lengths to contract. Unfortunately, the clocks of the 1890s were far too inaccurate to reveal the truth; and, faced with the scientific and technological triumphs of Newtonian physics, triumphs grounded firmly on the foundation of absolute time, nobody was willing to assert with conviction that time really does dilate. Lorentz, Poincaré, and Larmor waffled.

Einstein, as a student in Zurich, was not yet ready to tackle such heady issues as these, but already he was beginning to think about them. To his friend Mileva Marić (with whom romance was now budding) he wrote in August 1899, "I am more and more convinced that the electrodynamics of moving bodies, as presented today, is not correct." Over the next six years, as his powers as a physicist matured, he would ponder this issue and the reality of the contradiction of lengths and dilation of time.

Weber, by contrast, showed no interest in such speculative issues. He kept right on lecturing about Newtonian physics as though all were in perfect order, as though there were no hints of cracks in the foundation of physics.

As he neared the end of his studies at the ETH, Einstein naively assumed that, because he was intelligent and had not really done all that badly in his courses (overall mark of 4.91 out of 6.00), he would be offered the position of "Assistent" in physics at the ETH under Weber, and could use it in the usual manner as a springboard into the academic world. As an Assistent he could start doing research of his own, leading in a few years to a Ph.D. degree.

But such was not to be. Of the four students who passed their final exams in the combined physics mathematics program in August 1900, three got assistantships at the ETH working under mathematicians; the fourth, Einstein, got nothing. Weber hired as Assistents two engineering students rather than Einstein.